

The objective function is

$$f(x, y) = x^2 + 3y^2$$

Its gradient is

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 6y \end{pmatrix}$$

$$x^{(0)} = (5, 2), \quad v^{(0)} = (0, 0)$$

$$\alpha = 0.1, \quad \beta = 0.6$$

Using momentum convention from lecture 5, the updates are

$$v^{(k+1)} = \beta v^{(k)} + \nabla f(x^{(k)}), \quad x^{(k+1)} = x^{(k)} - \alpha v^{(k+1)}$$

$$k=0$$

The initial gradient is $\nabla f(5, 2)$, $v^{(0)} = (0, 0)$

$$\alpha = 0.1, \quad \beta = 0.6$$

$$v^{k+1} = \beta v^{(k)} + \nabla f(x^{(k)}),$$

$$x^{k+1} = x^{(k)} - \alpha v^{(k+1)}$$

Two momentum steps
iteration $k=0$

the initial gradient is

$$\nabla f(5,2) = (10, 12)$$

The initial velocity is $v^{(0)} = (0, 0)$

and the function value is

$$f(5,2) = 5^2 + 3 \cdot 2^2 = 37$$

$$k=1$$

$$v^{(1)} = 0,6 \cdot (0,0) + (10,12) = (10,12)$$

$$\text{The new point is } x^{(1)} = (5,2) - 0,1(10,12) = (4,08)$$

The function value is

$$f(4,0,8) = 4^2 + 3(0,8)^2 = 17,92$$

$$k=2$$

iteration $k=2$

$$\nabla f(4,0,8) = (8, 4,8)$$

$$v^2 = 0,6(10,12) + (8,4,8) = (14,12)$$

$$x^2 = (4,0,8) - 0,1(14,12) = (2,6, -0,4)$$

$$f(2,6, -0,4) = 2,6^2 + 3(-0,4)^2 = 7,24$$

$$X_{GD}^{(1)} = (5, 2) - 0,1(10, 12) = (4, 0,8)$$

$$X_{GD}^{(2)} = (4, 0,8) - 0,1(8, 4,8) = (3,2; 0,32)$$

$$f(X_{GD}^{(2)}) = 3,2^2 + 3(0,32)^2 = 10,5472$$

Method	point after 2 steps	function value
momentum	(2,6, -0,4)	7,24
gradient descent	(3,2, 0,32)	10,5472

to

Momentum in the valley

$$f(x_1, x_2) = (1-x_1)^2 + 100(x_2-x_1)^2$$

$$\nabla f(x) = \begin{pmatrix} 2(x_1-1) - 400x_1(x_2-x_1) \\ 200(x_2-x_1) \end{pmatrix}$$

β	iteration	Did f increase
0	32,076	No
0,5	16,014	Yes
0,8	6,329	Yes

0,9

3,020

Yes

0,95

1,324

Yes

0,99

3,185

Yes

Comparison on Q,

For the team's quadratic problem

$$f(x_1, x_2) = x_1^2 + 129x_2^2$$

the Hessian is $H = \begin{pmatrix} 2 & 0 \\ 0 & 258 \end{pmatrix}$

$$\lambda_{\min} = 2, \quad \lambda_{\max} = 258, \quad \kappa = \frac{258}{2} = 129$$

The initial point is $x^{(0)} = (1, 3, 0, 7)$

and the stopping rule is relative

$$\|\nabla f(x^{(k)})\| \leq 10^{-6} \|\nabla f(x^{(0)})\|$$

~~Classical heavy-ball parameters~~

~~The assignment specifies the classical~~

$$\alpha_{HB} = \frac{4}{(\sqrt{\lambda_{\max}} + \sqrt{\lambda_{\min}})^2}$$

$$\beta_{HB} = \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^2$$

~~0~~

$$\alpha_{HB} = \frac{4}{(\sqrt{258} + \sqrt{2})^2} \approx 0,0130362$$

$$\beta_{HB} = \left(\frac{\sqrt{129} - 1}{\sqrt{129} + 1} \right)^2 \approx 0,7025104$$

~~Comparison~~

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